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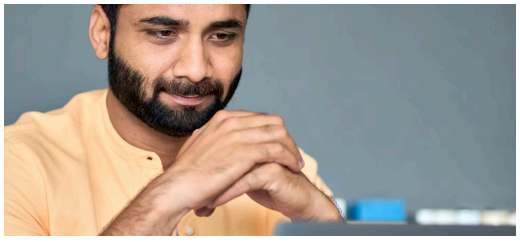
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Guide to Full-Stack Developer Skills

Written by Coursera Staff • Updated on Aug 26, 2025
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Gaining full-stack developer skills, including proficiency in programming languages and technologies, can open the door to becoming a full-stack developer. Discover how to become a full-stack developer and the skill set you need.



Full-stack developers have the skills to work on both the front and back ends of websites, video games, applications, and other projects. This career requires diverse skills like HTML, CSS, JavaScript, Oracle, SQL Server, Java, Python, Ruby on Rails, version control systems, project management, and more.

Below are the skills you need to pursue a career as a full-stack developer and the average salaries of full-stack developers in India.

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What is full-stack development?

A mobile app, website, or other development project is similar to a restaurant. The dining room's design aims to enhance customer comfort and encourage them to spend more time (and money) in the restaurant. But the kitchen's design is all about the chefs, helping them work as efficiently and effectively as possible. Both areas need thoughtful design, but the purposes behind the design are distinctly different.

A development project has a front-end, like the dining room of a restaurant, designed for the end user to experience the product. The back-end, like the kitchen of a restaurant, creates and stores all of the ingredients needed for the front-end to operate.

When the internet was new, developers commonly wrote entire programs by themselves. As technology advanced rapidly and computer programs became more complicated, developers started to specialise in one part of computer programming, like learning skills to design a website's user experience without knowing how to build the back-end or vice versa.

Today, the industry refers to a developer with the skills to manage a project's front-end, back-end, database, and DevOps as a "full-stack developer" who knows the "full-stack" of skills. Companies may experience many benefits when they hire a full-stack developer, and it's a lucrative career with growing demand in India.

What are the skills required for a full-stack developer?

You don't need a specific standard set of skills to become a full-stack developer, but you should learn basic front- and back-end skills to ensure you can work on diverse projects.

When deciding which skills to learn, you can refer to general guidelines to help you along the way, such as the MEAN stack, which stands for MongoDB, Express, Angular, and Node.js. These four technologies allow developers to call themselves "full-stack." Another example is the LAMP stack (Linux, Apache, MySQL, PHP/Perl/Python).

In general, a full-stack developer should be well-versed in two main areas of technology, which increases the scope of the skills you need to cultivate.

1. Front-end skills for full-stack developers

The front-end of a website or other development project refers to what the end user sees and interacts with. It's sometimes referred to as the presentation layer. Front-end developers impact product design and aesthetics.

Some programming languages you'll need as a front-end developer include:

- JavaScript/jQuery
- HTML
- CSS

A full-stack developer should have at least a working knowledge of these three languages, although you can learn many other technologies if you would like to specialise in front-end design. You also need to understand responsive design, which is the concept that a website or other project needs to be able to function on a range of devices, such as a laptop, smartphone, or tablet. Frameworks like Bootstrap or AngularJS are also great skills for a full-stack developer on the front-end.

2. Back-end skills for full-stack developers

The website's back-end is everything that goes into the website underneath the shiny top layer. This includes a server, database, and application. The back-end developer creates these ingredients so that the project functions.

Some of the common programming languages used for the back-end of a website include:

- Python
- Java
- Ruby
- Oracle
- SQL
- MySQL

A full-stack developer should be able to handle all aspects of web and application design, which requires proficiency in one or more server-side programming languages like Ruby or Python. Some other skills helpful for full-stack developers are PHP frameworks, Linux, and databases like SQL and MySQL.

You can choose to specialise in many areas of the back-end. Some of the biggest trends for full-stack developers are data analysis, machine learning and artificial intelligence, blockchain, mixed reality, and the Internet of Things (IoT).

3. General full-stack developer skills

In addition to front-end and back-end skills, a full-stack developer should have experience with a few general skills that all developers should develop. Developers are not usually working on projects by themselves, making communication and teamwork essential skills. Testing and debugging skills are also important for any developer to have, as well as version control, like GitHub India.

- Problem-solving
- Communication
- Testing/debugging
- Version control

- Teamwork
- Time management

What are the duties of a full-stack developer?

Due to their skills in both front-end and back-end development, the primary responsibility of a full-stack developer is to oversee all aspects of development projects, such as applications or websites, from the beginning stages to the end. Common job duties can include writing code, creating APIs, troubleshooting coding errors, updating software, and performing tests during the development process.



How to become a full-stack developer

You'll have many options once you know what skills you'd like to learn. You can pursue formal education, like a bachelor's degree in computer science, to learn many full-stack developer skills. You can also take individual courses and certification programmes to add to your credentials. Many boot camps are available to help you learn programming languages in a short period of intensive study. Lastly, consider job-readiness training, such as the Full Stack Software Developer Professional Certificate from IBM.

Full-stack developer average salary in India

The average salary for a full-stack developer in India is ₹7,24,670. Entry-level full-stack developers with less than one year of experience earn an average ₹4,04,000, while the average salary for an experienced full-stack developer can reach ₹2,000,000 [1].

Factors that affect salary include location, company size, and the applicant's skills. According to Payscale, a full-stack developer with Java can earn 27 per cent more than a developer who doesn't have that skill [1]. Microsoft Azure can help raise salaries by 83 per cent, while C# can help raise salaries by 24 per cent [1].

Why hire a full-stack developer?

Full-stack developers bring flexibility, reliability, and efficiency to a team. While a full-stack developer can only do the work of a part of a team, they can understand the work and see how it all comes together in the big picture.

In a team of developers where each professional manages or creates a different aspect of the project, you may experience pros and cons. Conversely, professionals likely deeply understand their part of the process. On the downside, being specialised in their area means they likely won't have a deep understanding of what's happening in the other silos of the project. This mismatch between departments could slow down development projects.

Bringing a full-stack developer onto the team means bringing in someone with a more extensive project picture. Having the skills to work on every project facet represents the ability to head off common problems by crossing silos in an organisation.

Enhance your full-stack developer skills on Coursera

Full-stack developers have an advantage in getting an education because you can take many different paths to develop a "full-stack" of developer skills.

If you want to become a full-stack developer and narrow down the number of education paths available, consider programmes that compile everything you need to know to get started. IBM offers a Full Stack Software Developer Professional Certificate on Coursera for individuals who need experience. A programme like that makes it easy to get started before you know exactly what you need to know.

If you want to take your skills and experience and round them out for a career as a full-stack developer, consider exploring the various options for Specialisations and Professional Certificates on Coursera. For example, Meta offers both a Front-End Developer and a Back-End Developer Professional Certificate, or you can find specialised offerings like IT Automation with Python Professional Certificate, provided by Google.



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Article sources

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